

Collisional stabilization of ion-molecule association complexes in He, H₂, or N₂ buffer gases

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ABSTRACT

The role of collisional energy transfer in ion-molecule association reactions is analyzed. Rate constants for the formation of adducts between HCl and NO₃-(HNO₃)_x (H₂SO₄)_y or HSO₄-(HNO₃)_x (H₂SO₄)_y (with x = 0–2 and y = 0–2) in the buffer gases H₂, He, and N₂ and at temperatures between 150 and 300 K are considered. Quantum-chemical calculations of molecular parameters and statistical unimolecular rate theory are combined to model low-pressure rate constants whereas ion-molecule capture theory provides high-pressure rate constants. The comparison with experimental results indicates that energy transfer is dominated by overall collision numbers while weak-collision effects are only of minor importance. On the other hand, often neglected falloff effects between termolecular and bimolecular reaction behavior have to be taken into account.

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1. Introduction

Association and recombination reactions in their limiting low-pressure, termolecular ranges, at steady state are characterized by partial equilibria between highly excited states of the adducts and separated reactants. In addition, their third-order rate constants are governed by collisional energy transfer, described by overall collision numbers *Z* and collision efficiencies β_c . This concept applies both to the association of neutrals and atoms [1], and to the corresponding reactions of ions with neutral molecules. In the latter case, however, the process often proceeds through one or more bound intermediates and it is of interest to what extent intramolecular vibrational redistribution (IVR) within the adducts competes with collisional stabilization. This has an effect on the shape of the falloff curves, i.e. the transition between limiting low-pressure, third-order, and limiting high-pressure, second-order, kinetics. Measurements of ion-molecule association rate constants, thus, can be used to gain insight into IVR in relation to collisional energy transfer.

Obviously it is difficult to separate the two contributions to the rate constants, i.e. the populations of the excited states of the adducts and the effective collisional energy transfer rate constants. It is, therefore, useful to take advantage of independent information from other sources. The rate of the primary association of an anion A[−] and a neutral molecule B, as well as the rate of IVR within the excited adduct AB^{−*}, can be characterized by statistical unimolecular rate theory, such as rigid-activated complex theory in the form of Rice-Ramsperger-Kassel-Marcus (RRKM) theory or the Statistical Adiabatic Channel Model (SACM) for loose-activated complex cases [2]. This includes the characterization of the relevant steady-state populations of the excited states of the adducts AB^{−*}. Whereas the collision numbers *Z* of neutral adducts AB* commonly are estimated by Lennard-Jones expressions, it appears more reasonable to characterize *Z* for ionic adducts AB^{−*} by ion-molecule capture theory, i.e. using the Langevin equation in the absence of a permanent dipole of the collider M or the Su-Chesnavich expression [3] in the presence of a dipole. Solutions of the master equation for collisions relate collision efficiencies β_c with average energies transferred per collision $\langle \Delta E \rangle$ [4]. There is evidence that, in contrast to the overall collision numbers *Z*, values for $\langle \Delta E \rangle$ in excited neutrals and ions

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under similar excitation conditions are of similar magnitude [5,6]. By applying the VENUS software [7], W. L. Hase has pioneered trajectory calculations of $\langle \Delta E \rangle$ for excited cationic adducts AB^{+*} [6].

While specifics of any give collision process may only be determined through direct experiment or calculation of the intermolecular potential, there are sufficient commonalities between collision processes that an appropriately informed generalized approach is possible. An avenue to decipher the effects of changing the physical nature of the buffer gas M is to measure association rate constants for a set of ions as functions of temperature and pressure in different M. Fortunately, such experiments were conducted several decades ago, but a detailed analysis of the buffer gas effects was never applied. In the present work we combine modern quantum-chemical calculations, determining excited state populations, with detailed statistical unimolecular rate theory [8] to investigate association reactions of common stratospheric anionic clusters in helium, hydrogen, and nitrogen buffer gases.

The earth's stratospheric ion chemistry is well-described by acid/base reactions to form stable core anions, followed by associations reactions to form clusters, and finally ligand-switching reactions. The resulting negative ion abundance is dominated by nitric and sulfuric acid clusters, i.e. $A^-(H_2SO_4)_x (HNO_3)_y$ where $A^- = HSO_4^-$ or NO_3^- . One of us has measured association rate constants forming clusters of these types with HCl over a broad temperature range and as a function of pressure [9,10]. Coarse interpretation of the data is straightforward: for a given cluster across buffer gases, the relative ratios of the association rate constant to the Langevin capture rate constant indicate the relative magnitudes of collision stabilization rates. However, a more detailed analysis appears fruitful.

While the association studies of [9,10] were performed at low temperatures (ranging from 300 down to 150 K), cluster dissociation reactions were investigated in Refs. [11,12] at higher temperatures (from 300 up to 600 K). As shown later, association reactions forming larger clusters in the experiments of [9,10] often were studied in the intermediate falloff range between the limiting low- and high-pressure ranges, such that particular attention had to be paid to falloff effects. As these effects depend on the temperature, a comparison of the low-temperature association and the high-temperature dissociation results also appears helpful.

A number of questions arise: foremost, does a statistical approach quantitatively describe the data throughout the temperature and pressure conditions sampled? If so, are the reactions occurring under limiting low-pressure collision conditions, where the association (and dissociation) rates are linear with pressure, under high-pressure collision conditions, where the rates do not depend on the pressure, or in the falloff region in between? To what extent can absolute values of $\langle \Delta E \rangle$ be extracted and how do these values vary across clusters and between buffers gases? Importantly, to what degree of precision can these values be determined and, relatedly, to what accuracy can the relevant cluster bond energies be determined, and how do these values compare to results of quantum-chemical calculations? The present article starts with a detailed description of the quantum-chemical methods used to calculate molecular parameters of the reactants and adducts, such as bond energies, vibrational frequencies, and rotational constants. Based on these results, excited state populations of the adducts are determined and implemented in rate calculations employing statistical unimolecular rate theory (in the version of [8]). A comparison with experimental results finally sheds light on the role of collisional processes and their competition with the intrinsic dynamics of the association complexes in a series of reactions with increasing adduct sizes.

2. Quantum-chemical calculations of molecular parameters

All quantum-chemical calculations were performed using the Gaussian software package [13]. For the adduct species HNO_3 , HNO_3-NO_3 , $(HNO_3)_2-NO_3$, H_2SO_4 , $H_2SO_4-HSO_4$, and $(H_2SO_4)_2-HSO_4$, geometry optimizations and frequency calculations were performed using several functionals (B3LYP [14–16], M06 [17], M062X [17], WB97XD [18], and PBE0 [19]) and basis sets (6–31++G (d,p), 6–311++G (d,p), def2-SVPD, def2-TZVPD). All basis sets include diffuse and polarization functions. Dispersion corrections were implemented using Grimme's DFT-D3 (GD3) method for each method except WB97XD, in which Grimme's DFT-D2 is included [20]. The bond energy E_0 (at temperatures $T = 0$ K, i.e. $E_0 = E_0(0\text{ K})$, and $T = 298$ K) for each cluster was calculated by the relationships

$$E_0(0\text{ K}) = \sum_{\text{prod}} (E_{\text{elec}} + \text{ZPE}) - \sum_{\text{react}} (E_{\text{elec}} + \text{ZPE}) + \text{BSSE} \quad (1)$$

$$E_0(298\text{ K}) = \sum_{\text{prod}} (E_{\text{elec}} + \text{ZPE} + H_{\text{corr}}) - \sum_{\text{react}} (E_{\text{elec}} + \text{ZPE} + H_{\text{corr}}) + \text{BSSE} \quad (2)$$

where E_{elec} is the electronic energy, ZPE is the zero-point energy, H_{corr} is the enthalpy correction at 298 K, and BSSE is the basis set superposition error calculated using a counterpoise (CP) correction. Relatively few reliable literature values exist for systems similar to those of interest here. Performance of different functionals and basis sets is compared to the proton affinities of NO_3^- and HSO_4^- , and to derived bond strengths of $NO_3-(HNO_3)$ and $HSO_4-(H_2SO_4)$. In addition, second-order Møller-Plesset (MP2) perturbation theory was used for several of the smaller clusters mentioned above. As shown below, the MP2 values for E_0 are generally in better agreement with the available experiments, but this method cannot be applied to larger systems due to its high calculational costs. The MP2 benchmark is used to select a functional-basis set combination for DFT calculations. Geometries across functionals were largely similar.

Calculated energetics showed only small differences between functionals and between basis sets. The M06 and WB97XD functionals showed marginally better performance on average, as did the def2-TZVPD basis set, although the differences are not sufficiently large to indicate a clear, systematic advantage. We select the M06 functional with the def2-TZVPD basis set for the larger cluster calculations discussed below. The optimized structures, energies, and harmonic frequencies are compared in the Supplementary Information (SI) for all combinations of methods and basis sets.

Using the M06/def2-TZVPD combination, the geometries of the HNO_3-HCl , $HNO_3-HCl-NO_3$, $(HNO_3)_2-HCl-NO_3$, H_2SO_4-HCl , $H_2SO_4-HCl-HSO_4$, $(H_2SO_4)_2-HCl-HSO_4$, and $HNO_3-HCl-HSO_4$ adducts were optimized with Grimme's DFT-D3 dispersion correction. In several systems, several convergent cluster geometries with similar energies for species involving HCl were identified. Optimized geometries of the nitric and sulfuric acid monomers in the DFT and MP2 calculations largely agree with convergent structures identified in previous theoretical studies [10,11,21]. For the cluster ions, there have been essentially no report of the geometries, so a comparison here is not possible.

The value for E_0 of $HNO_3 \rightarrow H^+ + NO_3^-$ (the proton affinity of NO_3^-) is calculated at several levels of DFT theory and compared to the experimental value (1352.31 ± 0.64 kJ mol $^{-1}$ at 0 K, 1357.53 ± 0.64 kJ mol $^{-1}$ at 298 K) [22]. The def2-TZVPD basis set provides good agreement (RMSE of 7 kJ mol $^{-1}$ across the functionals studied), while the other basis sets investigated systematically underestimate the proton affinity by 15–20 kJ mol $^{-1}$. The harmonic vibrational frequencies determined for nitric acid using the triple- ζ basis sets and the B3LYP, M06, M062X, and PBE0

methods exhibit scaling factors around 0.95 and reasonably good agreement with literature values [23,24]. For the $\text{H}^+ + \text{HSO}_4^- \rightarrow \text{H}_2\text{SO}_4$ system, all basis sets reasonably reproduce the experimental value [25] with RMSE across functionals of 5–10 kJ mol^{-1} . The BSSEs for the H_2SO_4 system, < 3.5 kJ mol^{-1} (<0.25% of the bond energy) for all levels of theory, were slightly larger than those for the nitrate system, and for both systems decreased with larger basis sets. Calculated harmonic frequencies for sulfuric acid also show good agreement with experimental values [26].

The geometries of the intermediate-sized anion clusters $\text{HNO}_3\text{--NO}_3^-$ and $\text{H}_2\text{SO}_4\text{--HSO}_4^-$ are largely similar to previously optimized results at the HF/6-31G(d) level [11,12] for all combinations of functionals and basis sets used in this work. The calculated bond energies of $\text{HNO}_3\text{--NO}_3^-$, however, are systematically lower than the reported value of $E_0 = 112 \text{ kJ mol}^{-1}$ [11] by around 20 kJ mol^{-1} . The calculated bond energies for $\text{H}_2\text{SO}_4\text{--HSO}_4^-$ show better agreement with the reported value $E_0 = 182 \text{ kJ mol}^{-1}$ [12], but are also systematically low by around 10 kJ mol^{-1} for all methods and basis set combinations, though the M06 functional produces the best results with all bond energies within 5 kJ mol^{-1} of the experimental value. The magnitude of the BSSE of the nitric acid-nitrate cluster is < 6 kJ mol^{-1} for all levels of theory and is less than 5% of the bond energy, while the BSSE of the sulfuric acid-sulfate system is larger (<17 kJ mol^{-1} for all levels of theory) and contributes < 9% to the total bond energy. For both systems, the dispersion correction results in lower bond energies that deviate further from the experimental values.

The optimized structures of the larger and mixed clusters $\text{NO}_3\text{--}(\text{HNO}_3)_2$, $\text{HSO}_4\text{--}(\text{H}_2\text{SO}_4)_2$, and $\text{HSO}_4\text{--HNO}_3$ exhibit geometries similar to those reported in previous studies [11,12]. The bond energy of the cluster $\text{NO}_3\text{--}(\text{HNO}_3)_2$ at the M06/def2tzvpd level of theory agrees with the experimental value $E_0 = 82 \text{ kJ mol}^{-1}$ [11] within 6 and 8 kJ mol^{-1} at 0 K and 298 K, respectively, while the $\text{HSO}_4\text{--}(\text{H}_2\text{SO}_4)_2$ cluster exhibits similar agreement to the reported value $E_0 = 120 \text{ kJ mol}^{-1}$ [12] deviating only 8 kJ mol^{-1} at both 0 K and 298 K. The mixed cluster $\text{HSO}_4\text{--HNO}_3$, which has an experimental bond energy of $E_0 = 114 \text{ kJ mol}^{-1}$ [12], exhibits bond energies lower by 14 kJ mol^{-1} and 15 kJ mol^{-1} at 0 K and 298 K, respectively, when calculated at the M06/def2-TZVPD level of theory. The BSSEs of $\text{NO}_3\text{--}(\text{HNO}_3)_2$ and $\text{HSO}_4\text{--HNO}_3$ account for less than 2% of the total bond energy, while the BSSE for $\text{HSO}_4\text{--}(\text{H}_2\text{SO}_4)_2$ is greater in magnitude and accounts for 7% of the total bond energy (see Table 1).

Multiple convergent configurations were identified for many of the cluster species that include an HCl moiety, and the bond energy of each isomer is contained in Table 2. The two isomers of $\text{NO}_3\text{--HNO}_3\text{--HCl}$ appear structurally and energetically similar with

Table 1
Calculated bond energies (M06/def2-TZVPD) E_0 (kJ mol^{-1}) for lowest energy isomers of each indicated species. Full results for all isomers available in SI.

Species	E_0 (0 K)	E_0 (298 K)
$\text{H}^+\text{--NO}_3^-$	1355	1361
$\text{NO}_3\text{--HNO}_3$	124	124
$\text{NO}_3(\text{HNO}_3)\text{--}(\text{HNO}_3)$	77	75
$\text{H}^+\text{--HSO}_4^-$	1307	1313
$\text{HSO}_4\text{--H}_2\text{SO}_4$	186	189
$\text{HSO}_4(\text{H}_2\text{SO}_4)\text{--H}_2\text{SO}_4$	112	112
$\text{HSO}_4\text{--HNO}_3$	128	129
$\text{NO}_3\text{--HCl}$	94	97
$\text{NO}_3\text{HNO}_3\text{--HCl}$	44	46
$\text{NO}_3(\text{HNO}_3)_2\text{--HCl}$	28	30
$\text{HSO}_4\text{--HCl}$	99	104
$\text{HSO}_4\text{H}_2\text{SO}_4\text{--HCl}$	37	38
$\text{HSO}_4(\text{H}_2\text{SO}_4)_2\text{--HCl}$	29	30
$\text{HSO}_4\text{HNO}_3\text{--HCl}$	42	44

Table 2

Experimental and calculated limiting strong-collision, low-pressure rate constants (represented in third-order form $k/[M]$ in $10^{-27} \text{ cm}^6 \text{ s}^{-1}$).

Reaction	T	$k_{\text{exp}}/[M]$	$k_{0,\text{calc}}/[M]$
$\text{NO}_3^- + \text{HCl} + \text{N}_2$	152	3.0	3.5
	161	2.5	3.0
	172	2.2	2.4
	224	1.1	1.1
$\text{NO}_3^- + \text{HCl} + \text{H}_2$	152	3.9	7.9
	171	2.7	5.6
	227	1.7	2.4
	295	0.95	1.1
$\text{NO}_3^- + \text{HCl} + \text{He}$	152	2.5	2.9
	171	1.5	2.0
	200	1.0	1.3
	225	0.63	0.91
$\text{HSO}_4^- + \text{HCl} + \text{N}_2$	237	0.46	0.78
	294	0.26	0.41
	150	16	18
	187	9.3	7.7
$\text{HSO}_4^- + \text{HCl} + \text{H}_2$	228	4.7	3.4
	149	13	43
	187	6.6	18
	229	3.6	7.8
$\text{HSO}_4^- + \text{HCl} + \text{He}$	149	6.5	15
	187	3.0	6.5
	230	1.4	2.8

bond energies of 44–47 kJ mol^{-1} that differ by $\leq 1 \text{ kJ mol}^{-1}$ at 0 K and 298 K, and the harmonic frequencies also show slight differences. Three distinct isomers were identified for $\text{NO}_3\text{--}(\text{HNO}_3)_2\text{--HCl}$, two having very similar bond energies of 28–31 kJ mol^{-1} with $\leq 1 \text{ kJ mol}^{-1}$ differences and the other having bond energies 31–33 kJ mol^{-1} , only 2 kJ mol^{-1} lower than the other two. One low-lying isomer with bond energy 31–33 kJ mol^{-1} was identified for the $\text{HSO}_4\text{--H}_2\text{SO}_4\text{--HCl}$ system as well as two other distinct isomers at +5 and +8 kJ mol^{-1} , while two low-lying isomers were found for the $\text{HSO}_4\text{--}(\text{H}_2\text{SO}_4)_2\text{--HCl}$ system, which also exhibited two other isomers, each at +5 kJ mol^{-1} . The two isomers of $\text{HSO}_4\text{--HNO}_3\text{--HCl}$ are energetically similar within 0.01 kJ mol^{-1} of each other and have almost identical harmonic frequencies suggesting they are structurally equivalent.

In general, calculated bond energies between different clusters generally agree with available experimental values within 15–20 kJ mol^{-1} , an uncertainty typical of 2σ uncertainty for DFT methods [21]. We note that uncertainties in calculated energies are most commonly reported as mean - average deviations, equivalent to $\sim 0.75\sigma$ assuming the errors are normally distributed, or root mean squared deviation, equivalent to $\sim \sigma$, and that these values should be multiplied by ~ 2.5 or ~ 2 respectively to achieve the 2σ uncertainty most commonly reported in experimental studies. Double- ζ basis sets lead to larger errors in the bond energies for cases where the smaller basis sets result in calculated bond energies that are lower than the experimental values. The inclusion of the dispersion corrections also have a large effect on the bond energy. On the other hand, the BSSE error is typically quite small, constituting less than 1% of the bond energy.

3. Statistical calculation of rate constants

We calculate the rate constants k for ion-molecule association reactions in the framework of statistical unimolecular rate theory, e.g. in the formulation of [8]. First-order dissociation rate constants k_{uni} here are converted into second-order association rate constants k with the help of the equilibrium constants K_c . The latter are given by

$$k_c = k_{\text{uni}}/k = [Q(A^-)Q(B) \exp(-E_0/k_B T)]/Q(AB^-) \quad (3)$$

with translational, rotational, and vibrational contributions in the respective partition functions Q and with the A^- - B bond energy E_0 at 0 K (we assume that only single electronic states contribute). In the treatment of [8], the limiting low-pressure rate constant $k_{\text{uni},0}$ is factorized in the form

$$k_{\text{uni},0} \approx [M] \beta_c Z [\rho_{\text{vib},h}(E_0) k_B T / Q_{\text{vib}}(AB^-)] \exp(-E_0/k_B T) F_{\text{anh}} F_E F_{\text{rot}} F_{\text{rotint}} \quad (4)$$

(for the detailed meaning of the contributing factors, see Ref. [8]; it may be mentioned that eq. (4) was also used in the analysis of low-pressure dissociation results of [11,12]). One notes that the Boltzmann factors $\exp(-E_0/k_B T)$ (as well as the vibrational partition functions $Q_{\text{vib}}(AB^-)$ of the adducts) cancel in the conversion of the low-pressure dissociation rate constants $k_{\text{uni},0}$ into the low-pressure association rate constants k_0 . Nevertheless, the bond energies E_0 of the adducts enter the expression for k_0 as well as for $k_{\text{uni},0}$ through the harmonic vibrational density of states $\rho_{\text{vib},h}(E_0)$ of the excited adducts (together with other factors in eq. (4), in particular F_{rot}). Eqs. (3) and (4), therefore, provide a valuable access to the relevant E_0 , once k_0 is measured and the molecular parameters entering $\rho_{\text{vib},h}(E_0)$ have been calculated. A comparison with calculated E_0 then can be made. This is an important aspect of the present article.

A few details of our rate calculations are mentioned in the following: $\rho_{\text{vib},h}(E_0)$ of the adducts are determined employing the Beyer-Swinehart state counting routine and represented in the Whitten-Rabinovitch form; anharmonicity factors F_{anh} and energy dependence factors F_E are estimated as described in Ref. [8]. Rotational factors F_{rot} are expressed by their maximum values F_{rotmax} neglecting centrifugal barriers; internal rotation factors F_{rotint} are used only when internal rotations contribute in addition to vibrations.

Overall collision numbers Z for the present buffer gases He, H_2 , and N_2 are estimated by the Langevin equation

$$Z \approx 2\pi q(\alpha/\mu)^{1/2} \quad (5)$$

with the ionic charge q , the polarizability α of M and the reduced mass μ of the $AB^- + M$ collision pair. (quadrupole contributions of H_2 and N_2 to Z are neglected [27]). According to the relationship [4],

$$\beta_c / (1 - \beta_c^{1/2}) \approx - \langle \Delta E \rangle / (F_E k_B T) \quad (6)$$

experimental data for β_c provide an access to $\langle \Delta E \rangle$ only as long as β_c is noticeably smaller than unity. However, any conclusion on $\langle \Delta E \rangle$ by eq. (6) from an apparent β_c , which is smaller than unity, is justified only when the experimental data are taken close to the low-pressure limit of the reaction. As sufficiently large pressure ranges in the experiments often were not accessible, it is necessary to estimate the limiting high-pressure association rate constant k_∞ . This is done by the use of the Su-Chesnavich relationship [3] for capture of A^- by B . The concentration $[M]$, for which $k_0 = k_\infty$ defines the “center $[M]_{\text{cent}}$ of the falloff curve”. One should remember that the second-order rate constant k_0 is proportional to $[M]$ (for the same reaction system and under the same experimental conditions, $[M]_{\text{cent}}$ is identical for dissociation and the reverse association). Only when the experimental $[M]$ is much smaller than $[M]_{\text{cent}}$, an apparent lowering of k to values below k_0 due to falloff effects [28,29] can be ruled out. One should also remember that the position of falloff curves along the buffer gas pressure scale shifts with temperature, i.e. $[M]_{\text{cent}}$ depends on the temperature. Values

of $[M]_{\text{cent}}$ in the association experiments of [9,10] for similar buffer gas pressures correspondingly are expected to be smaller than the values of the dissociation experiments of [11,12]. Falloff-lowering effects of k can be expressed in terms of the ratio $k([M]_{\text{cent}})/k_0$ ($[M]_{\text{cent}} = F_{\text{cent}}/2$ where F_{cent} is termed “center broadening factor” of the falloff curve. Simple estimates of F_{cent} can be made with the methods suggested in Refs. [28,29].

4. Results and discussion

4.1. Association reactions $\text{NO}_3 \cdot (\text{HNO}_3)_n + \text{HCl} + M \rightarrow \text{NO}_3 \cdot (\text{HNO}_3)_n(\text{HCl}) + M$

The association reactions $\text{NO}_3 \cdot (\text{HNO}_3)_n + \text{HCl} + M \rightarrow \text{NO}_3 \cdot (\text{HNO}_3)_n(\text{HCl}) + M$ for $n = 0, 1$, and 2 have been studied experimentally in the buffer gases $M = \text{He}, H_2$, and N_2 over the temperature range 152–295 K [9]. H_2 and N_2 buffer gas pressures near 0.15 Torr were employed (at 150 K corresponding to about $[M] = 9.7 \times 10^{15} \text{ cm}^{-3}$; He pressures were near 0.4 Torr). Table 2 shows the measured rate constants, expressed in the form of third-order rate constants $k/[M]$.

One first asks how close the experimental conditions were to the limiting low-pressure range of the association reactions. This question is answered by comparing the measured second-order rate constants k at the employed $[M]$ with second-order high-pressure rate constants k_∞ . The latter (near 150 K) can be estimated by the Su-Chesnavich expression [3] to be near to $1.8 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ for $n = 0$ (or 1.6×10^{-9} for $n = 1$ and 1.5×10^{-9} for $n = 2$). As a consequence, the calculated $[M]_{\text{cent}}$ was of the order of $5 \times 10^{17} \text{ cm}^{-3}$ (at 150 K) for $n = 0$ such that a falloff reduction of k to values below k_0 here should have been smaller than the experimental uncertainty. The situation changes for $n = 1$ where $[M]_{\text{cent}}$ is calculated to be about ten times smaller than for $n = 0$ such that, due to the effects of increasing broadening of the falloff curves [28,29], the experimental k should have been somewhat below the corresponding k_0 , although deviations from third-order behavior would have been difficult to detect experimentally (similar deviations had to be expected for $n = 2$). A more detailed modelling of the falloff curves would be required to arrive at more quantitative conclusions, but this is outside the scope of the present article.

Assuming that there were no major falloff problems for $n = 0$, the performance of the rate modelling can best be tested for this case, i.e. for the reaction $\text{NO}_3 \cdot + \text{HCl} + M \rightarrow \text{NO}_3 \cdot (\text{HCl}) + M$. Assuming that anharmonicity effects remain small ($F_{\text{anh}} \approx 1$), we first calculate strong-collision rate constants ($\beta_c = 1$) for $M = N_2$. The resulting values are given in Table 2. The calculated values are in remarkable agreement with the experimental values, both in absolute value and temperature dependence. This may indicate that N_2 in this case is almost a strong collider, with $-\langle \Delta E \rangle / (F_E k_B T)$ being larger than unity (F_E here is not far from unity). However, experimental and modelling uncertainties do not allow to be more quantitative at this stage. Modelling of strong-collision rate constants for $M = H_2$ and He and comparison with the experimental values in Table 2 leads to agreement within about a factor of two, although the calculated strong-collision rate constants were systematically somewhat larger than the experimental values (apparent weak collision efficiencies β_c between 0.5 and unity). According to eq. (6) these effects could be attributed to values of $-\langle \Delta E \rangle / (F_E k_B T)$ of the order of unity. However, the effects appear too small, to allow for more quantitative conclusions on $\langle \Delta E \rangle$. Buffer gas effects under these conditions instead most probably are not dominated by differences in $\langle \Delta E \rangle$ but in Z . The good agreement of the calculated and measured low-pressure rate constants k_0 for $M = N_2$ is illustrated in Fig. 1.

The good performance of the statistical modelling of k_0 for $n = 0$

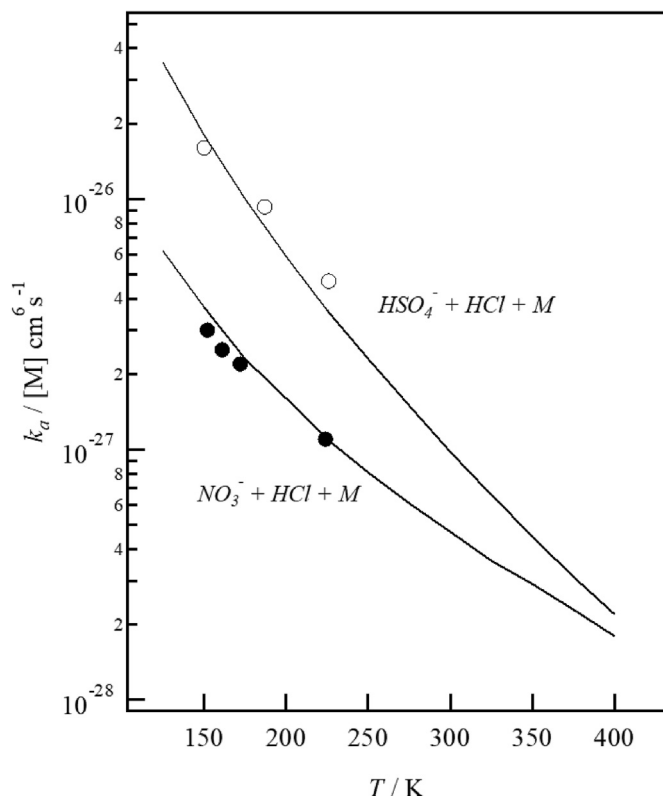


Fig. 1. Limiting low-pressure rate constants $k_0/[N_2]$ for the association reactions of HCl with NO_3^- and HSO_4^- in the buffer gas N_2 (full lines: modelled strong-collision rate constants; circles: experimental rate constants from Refs. [9,10], filled circles for $HCl + NO_3^-$, open circles for $HCl + HSO_4^-$).

suggests that the approach can be used as a diagnostic for the interpretation of the experimental results for larger clusters, i.e. for $n = 1$ and 2. Again we calculated strong-collision rate constants (i.e. $\beta_c = 1$). The results within about a factor of two again agree with the measurements for $M = He$ and H_2 , suggesting that weak-collision effects (i.e. $\beta_c < 1$) were not particularly pronounced and that falloff reductions of the measured rate constants k to values below k_0 were not detectable.

4.2. Association reactions $HSO_4^-(H_2SO_4)_m(HNO_3)_n + HCl + M \rightarrow HSO_4^-(H_2SO_4)_m(HNO_3)_n(HCl) + M$

Clustering of HCl with $HSO_4^-(H_2SO_4)_m(HNO_3)_n$ was investigated experimentally for $n = 0$ and 1 and $m = 0-3$ in Ref. [10]. Likewise, the association of NO_3^- with HNO_3 was studied in the buffer gas $M = He$. The latter experiments over the pressure range 0.15–1 Torr clearly showed “saturation effects”, i.e. deviations from limiting low-pressure behavior. Estimating k_∞ again by the Su-Chesnavich equation, at 300 K leads to $k_\infty \approx 2.7 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$, indicating that the experimental value of $2.5 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ at 1 Torr was still far below k_∞ and only part of the broad falloff curve had been observed. This signals the importance of falloff reductions of the rate constants k to values below k_0 in the only assumed termolecular range of the reaction. With the molecular parameters given above and in the SI, with the formalism of [28,29] one estimates F_{cent} to be of the order of 0.05. This explains the magnitude of the falloff reduction of k to values below k_0 . Even more pronounced falloff reductions, therefore, should be expected for larger clusters such as $HSO_4^-(H_2SO_4)_m(HNO_3)_n(HCl)$ with $m > 0$.

The clustering of HCl with HSO_4^- under the experimental

conditions of [10] turned out to be studied close to the limiting low-pressure range of the association reaction. In this case, k_∞ for capture of HCl by HSO_4^- is estimated to be $1.6 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$ such that $[M]_{cent}$ (for $M = N_2$ and 150 K) of the order of 10^{17} cm^{-3} apparently was sufficiently above the experimental $[M]$ (of the order of 10^{16} cm^{-3}). Modelling of strong-collision rate constants like for $HCl + NO_3^- + M$ again led to reasonable agreement between modelled and measured k_0 such as shown in Table 2. The situation changes for larger clusters. As an example we consider the clustering $(HSO_4^-)(HNO_3) + HCl + M$. With $k_\infty \approx 1.5 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$, in this case $[M]_{cent}$ is of the order $5 \times 10^{16} \text{ cm}^{-3}$. As the falloff curve in this case will be broader than for the smaller cluster [28,29], one should observe falloff reductions of the apparent low-pressure rate constants. Such reductions indeed are observed when the size of the association cluster increases, see Table 3. The rate modelling in this case accounted only for the population of one isomer (named 1 in the SI). Including populations of more isomers, which at low pressures should have enough time to randomize with the primary isomer 1, would place $[M]_{cent}$ at even smaller values, further broadening the falloff curve and increasing the extent of falloff reduction of k . Falloff reductions of k to values below k_0 should be even more pronounced for larger clusters which are not further analyzed here.

5. Conclusions

On the basis of quantum-chemical calculations of molecular parameters and rate constants calculated by statistical unimolecular rate theory, limiting low-pressure rate constants k_0 for a series of cluster association reactions were determined. In addition, limiting high-pressure rate constants k_∞ were estimated by ion-molecule capture theory. The comparison with experimental results indicates that most of the reactions forming larger clusters in the experiments were investigated under conditions corresponding to the intermediate falloff range of the reaction. The initial assumption that all of these reactions were studied as termolecular processes in their limiting low-pressure range, therefore, needs to be revised. Only for small adducts, like $NO_3^-(HNO_3)$, $NO_3^-(HCl)$, and $HSO_4^-(HCl)$, the rate constants k were close to k_0 . The comparison of calculated strong-collision rate constants k_0 and the experimental values for these cases suggests that collisional effects were dominated by overall collision numbers Z , such as estimated by the Langevin collision number, and collision efficiencies β_c smaller than unity were of only minor importance. The apparent values of β_c were found to be systematically somewhat larger for the buffer gas N_2 than for the buffer gases H_2 and He , but only lower limits of the ratios $-\langle \Delta E \rangle / (E_{\text{rel}} k_B T)$ (being of the order of unity) could be extracted from the measurements. The importance of falloff reductions of measured k to values below the third-order limiting values k_0 has been stressed for the formation of larger association adducts. The accuracies of molecular parameters (well-depths, vibrational and rotational frequencies) calculated using density functional methods were sufficient to enable the modeling, but

Table 3

Experimental and calculated limiting strong-collision, low-pressure rate constants at $150 \pm 2 \text{ K}$ (represented in third-order form $k/[M]$ in $10^{-27} \text{ cm}^3 \text{ s}^{-1}$).

Reaction	$k_{\text{exp}}/[M]$	$k_{0,\text{calc}}/[M]$
$NO_3^- + HCl + H_2$	3.9	7.9
$NO_3^-(HNO_3) + HCl + H_2$	43	83
$NO_3^-(HNO_3)_2 + HCl + H_2$	23	15
$HSO_4^- + HCl + H_2$	13	43
$HSO_4^-(HNO_3) + HCl + H_2$	20	160
$HSO_4^-(H_2SO_4) + HCl + H_2$	11	81

insufficient for a more quantitative analysis of the minor contributions of weak-collision effects.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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